Sparse Biomarker Discovery of tDCS Treatment Response in Adults with Treatment-Resistant Depression using Low Montage Resting-State EEG

Abin Daniel Zorto[1]*, Jijomon C. Moncy[2],
Mhd Saeed Sharif[1], Cynthia H.Y. Fu[2]

[1]Intelligent Technologies Research Group, University of East London, University Way, London E16 2RD, UK

[2]School of Psychology, University of East London, University Way, London E16 2RD, UK

u2091940@uel.ac.uk, S.Sharif@uel.ac.uk,
c.fu@uel.ac.uk

Abstract

We evaluated whether a sparse, low-montage resting-state EEG pipeline could identify clinically interpretable candidate biomarkers of response to transcranial direct current stimulation (tDCS) in adults with treatment-resistant depression under strict patient-level validation. The study used 4-channel eyes-closed EEG and a controlled sweep of 150 successful runs across two model families: an advanced hybrid 1D CNN-LSTM and a linear support vector machine (SVM). The sweep varied window size (2–10 s) and the per-fold feature-selection budget (`inner-k` 1–70), while holding the consensus feature budget fixed at 10 and enabling both leave-one-participant-out group equalization and SMOTE during training.

The strongest configuration was the hybrid model with 6 s windows and `inner-k=1`. That run reached patient-level ROC-AUC 0.816, PR-AUC 0.579, balanced accuracy 0.786, accuracy 0.810, and MCC 0.571 over 21 held-out participants. The best SVM configuration (8 s, `inner-k=30`) reached ROC-AUC 0.510 and MCC 0.277. The hybrid winner was also highly sparse, selecting 1 feature per fold and only 7 unique features overall, with a permutation-tested ROC-AUC advantage over chance ($p = 0.008$). Hard-set overlap across folds was modest (mean pairwise Jaccard 0.190; Kuncheva index 0.187). The recurrent features showed perfect mean effect-direction consistency.

These results support a sparse short-timescale EEG framework that recovers a small, interpretable candidate biomarker set while preserving useful patient-level discrimination in a relatively small cohort. The most recurrent features involved temporal spectral entropy and frontal-temporal asymmetry-derived measures, and their class-conditional selection patterns motivate follow-up validation in larger externally evaluated datasets.

1 Introduction

Treatment-resistant depression (TRD) remains a major clinical challenge, and neuromodulation approaches such as transcranial direct current stimulation (tDCS) continue to attract interest as potentially scalable adjunctive interventions (Milev et al., 2016; Boggio et al., 2007; ?; ?; ?; Thair et al., 2017; Cramer et al., 2011). More recent work has also pushed tDCS into remotely supervised

*Corresponding author: Abin Daniel Zorto, u2091940@uel.ac.uk

and home-based settings, which makes pretreatment stratification a practical clinical problem (??). A useful biomarker pipeline has to identify which patients are more or less likely to respond before stimulation begins.

In that setting, EEG is appealing because it is comparatively inexpensive, portable, and already used across translational neurophysiology workflows. Wearable and reduced-channel EEG systems further increase deployment feasibility, although they also sharpen the trade-off between practicality and signal richness (??BABILONI et al., 2012; ?; Simmatis et al., 2023; ?).

The literature on EEG biomarkers in major depressive disorder is encouraging and methodologically uneven. Reviews and biomarker overviews have reported potentially informative spectral, asymmetry, vigilance, entropy, and complexity signatures, while also emphasizing cohort heterogeneity, protocol sensitivity, and the need for clinically useful predictive markers (Olbrich and Arns, 2013; Olbrich et al., 2015; Leuchter et al., 2010; ?; ?; Simmatis et al., 2023; ?; ?). Machine learning studies have extended this theme into antidepressant, placebo, and neuromodulation response prediction. External generalization, feature stability, and interpretability remain active concerns (?Jaworska et al., 2019; Oakley et al., 2022; Shalbaf et al., 2018; Ebrahimzadeh et al., 2021; ?; ?; ?; ?).

That tension is especially sharp in relatively small neuromodulation studies. Rich models can improve representation learning, but relatively small cohorts also increase the risk of unstable feature selection, inflated performance estimates, and overly confident biomarker claims. Clinically deployable EEG biomarkers also need interpretability and low-density feasibility. Emerging response-prediction studies in depression and related neuromodulation settings show substantial promise alongside clear methodological fragility (Funk and George, 2008; Bailey et al., 2023; Murphy et al., 2023; Guo et al., 2024; Tsai et al., 2023; ?; ?; ?; ?; Sheen et al., 2024; Zeidabadi and Rashidi, 2025; Reissmann et al., 2026; Ulrich et al., 2025). Within the same home-based 4-channel MDD program studied here, recent PSD-based and PLV-based analyses have already shown that the baseline EEG carries treatment-response information across multiple signal representations (??). That background makes this cohort a useful setting for testing whether a sparse interpretable feature-selection workflow can isolate a small recurrent biomarker set under strict participant-level validation.

This manuscript focuses on a narrow objective: sparse biomarker discovery under strict participant-level evaluation. We compare two classifiers over the same feature-selection sweep to test whether a low-montage resting-state EEG pipeline can recover a small recurring set of candidate markers associated with tDCS response in adults with TRD. The emphasis is on parsimony, candidate-biomarker interpretability, and leakage-resistant validation. Our working hypothesis was that useful patient-level discrimination could be achieved with a small per-fold feature budget and that the most recurrent features would show stronger effect-direction consistency than hard-set overlap.

2 Methods

2.1 Dataset and signal representation

The analysis used pretreatment EEG from 21 adults with treatment-resistant depression enrolled in a fully remote, multisite, double-blind, randomized sham-controlled home-based tDCS trial for major depressive disorder (?). The analyzed cohort contained 7 remission cases and 14 non-remission cases in the patient-level outcome labels, with 18 female participants and a mean age of 37.1 ± 9.7 years. Table 1 summarizes the cohort, acquisition, and sweep configuration.

Table 1: Cohort, acquisition, and sweep configuration summary. Trial descriptors derive from the parent home-based tDCS study (**?**); acquisition totals and sweep settings derive from the analyzed EEG exports and current experiment configuration.

Characteristic	Value
Parent trial and cohort descriptors	
Parent trial	Fully remote, multisite, sham-controlled home-based tDCS trial for MDD (**?**)
Participants	21
Outcome groups	7 remission, 14 non-remission
Sex (female/male)	18/3
Age (years, mean ± SD)	37.1 ± 9.7
EEG acquisition and export descriptors	
Rest condition	Eyes-closed resting state
EEG montage	4 channels (AF7, AF8, TP9, TP10)
Channel coverage	Frontal (AF7, AF8), temporal (TP9, TP10)
Sampling rate	256 Hz
Recording structure	10 min total; single file or two 5 min files
Available export segment length	10 s (2,560 samples/channel)
Total exported windows	1,203
Remission windows	393 (32.7%)
Non-remission windows	810 (67.3%)
Class ratio	2.06:1 (non-remission:remission)
Current sweep methodology settings	
Sweep window sizes	2, 4, 6, 8, 10 s
Inner-k sweep	1, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70
Outer-k	10
Feature selector	SelectKBest (f-classif)
Training adjustments	LOPO-group equalization + SMOTE
Validation	Leave-one-participant-out
Evaluation level	Patient-level aggregation

Baseline recordings were collected during eyes-closed rest with a 4-channel montage (`AF7`, `AF8`, `TP9`, `TP10`) sampled at 256 Hz. The available study exports comprised 10 s segments with 2,560 samples per channel, and the underlying recording sessions were stored either as a single 10 min file or as two 5 min files depending on participant. Across the cohort, the export set contained 1,203 windows, including 393 remission-labeled and 810 non-remission-labeled windows. This acquisition setup uses a limited frontal-temporal montage with clear portable-deployment relevance (**??**). The montage also constrained the downstream feature space toward frontal, temporal, cross-hemispheric, and frontal-temporal interaction measures.

For the present sweep, the processing configuration applied participant-level per-channel de-meaning before upsampling, followed by a fourth-order Butterworth low-pass filter at 60 Hz, a downsampling step back to 256 Hz, and zero-overlap window slicing. The sweep then compared non-overlapping analysis windows of 2, 4, 6, 8, and 10 s, so the model inputs included both the native 10 s exports and shorter subdivisions derived from the same recordings. These choices materially affect downstream EEG feature distributions, especially in low-density or portable pipelines (**???**BABILONI et al., 2012; Simmatis et al., 2023). The feature extractor enabled spectral, temporal, entropy, complexity, connectivity, coherence, asymmetry, and cross-hemispheric descriptors, including frontal/temporal asymmetry, cross-regional ratios, and inter-channel coupling measures.

This broad handcrafted feature pool was intended to preserve interpretability while allowing sparse downstream selection.

2.2 Sweep design and validation

The experiment runner launched two model families over the same feature-selection sweep: an advanced hybrid 1D CNN-LSTM and a linear SVM. All runs used sequential window ordering, SelectKBest with ANOVA F statistics (`select_k_best_f_classif`), a fixed consensus feature budget of `outer-k=10`, leave-one-participant-out outer evaluation, leave-one-participant-out training-group equalization, and SMOTE within training folds. The sweep varied the per-fold feature-selection budget (`inner-k`) across 15 values from 1 to 70 and repeated the search across 5 window sizes, yielding 150 successful runs in total (75 per model).

This design matters for interpretation. The outer evaluation remained subject independent throughout, so no participant contributed both training and test windows to the same outer fold. The `inner-k` parameter controlled how many features were selected inside each outer training fold, whereas `outer-k` controlled only the final consensus feature budget after cross-validation. The feature-selection stability panels in the manuscript describe shared pipeline behavior because both classifiers consumed the same selected features for matched sweep settings.

2.3 Models

The SVM baseline used a linear kernel with class-balanced weighting and probability output enabled. The hybrid model used a substantially richer architecture defined in the project model configuration: stacked convolutional blocks, bidirectional LSTM layers, multi-head attention with positional encoding, feature-pyramid fusion, and GELU-activated dense layers with dropout and batch normalization. The manuscript treats this architecture as a test case for whether a more expressive learner can exploit a small selected feature set more effectively than a linear baseline.

2.4 Outcome metrics and biomarker summaries

The primary endpoint was patient-level ROC-AUC, computed from participant probabilities aggregated over held-out windows. Secondary patient-level metrics included PR-AUC, balanced accuracy, accuracy, F1, MCC, precision, recall, and specificity. The pipeline also reported bootstrap confidence intervals and a permutation test for the patient-level ROC-AUC summary. All main manuscript claims use patient-level metrics. Window-level scores appear only as intermediate outputs.

To characterize biomarker behavior across outer folds, we used several post hoc summaries: mean pairwise Jaccard overlap, Kuncheva index, unique feature count, top-feature share, class-conditional selection deltas, and effect-direction consistency. These summaries are descriptive. They characterize recurrence and directionality across folds and do not support causal interpretation. Low overlap can coexist with recurring features that retain consistent class-separation direction when selected (Olbrich and Arns, 2013; Olbrich et al., 2015; Simmatis et al., 2023).

3 Results

3.1 Sweep-level comparison

Across the 150 successful runs, the hybrid model outperformed the SVM baseline on the main patient-level discrimination metrics over most of the sweep landscape (Figure 1). The performance advantage was visible both when averaging over window sizes and when averaging over `inner-k` settings. The best individual hybrid run occurred at 6 s windows with `inner-k=1`, whereas the best SVM run occurred at 8 s windows with `inner-k=30`.

The lower half of Figure 1 highlights a second, and more important, pattern. Increasing `inner-k` increased both hard-set overlap and the total number of unique selected features. Larger feature budgets did not produce the strongest patient-level discrimination. The sweep concentrated its best performance at the sparsest end of the per-fold selection range. This empirical pattern motivates the paper's focus on sparse biomarker discovery.

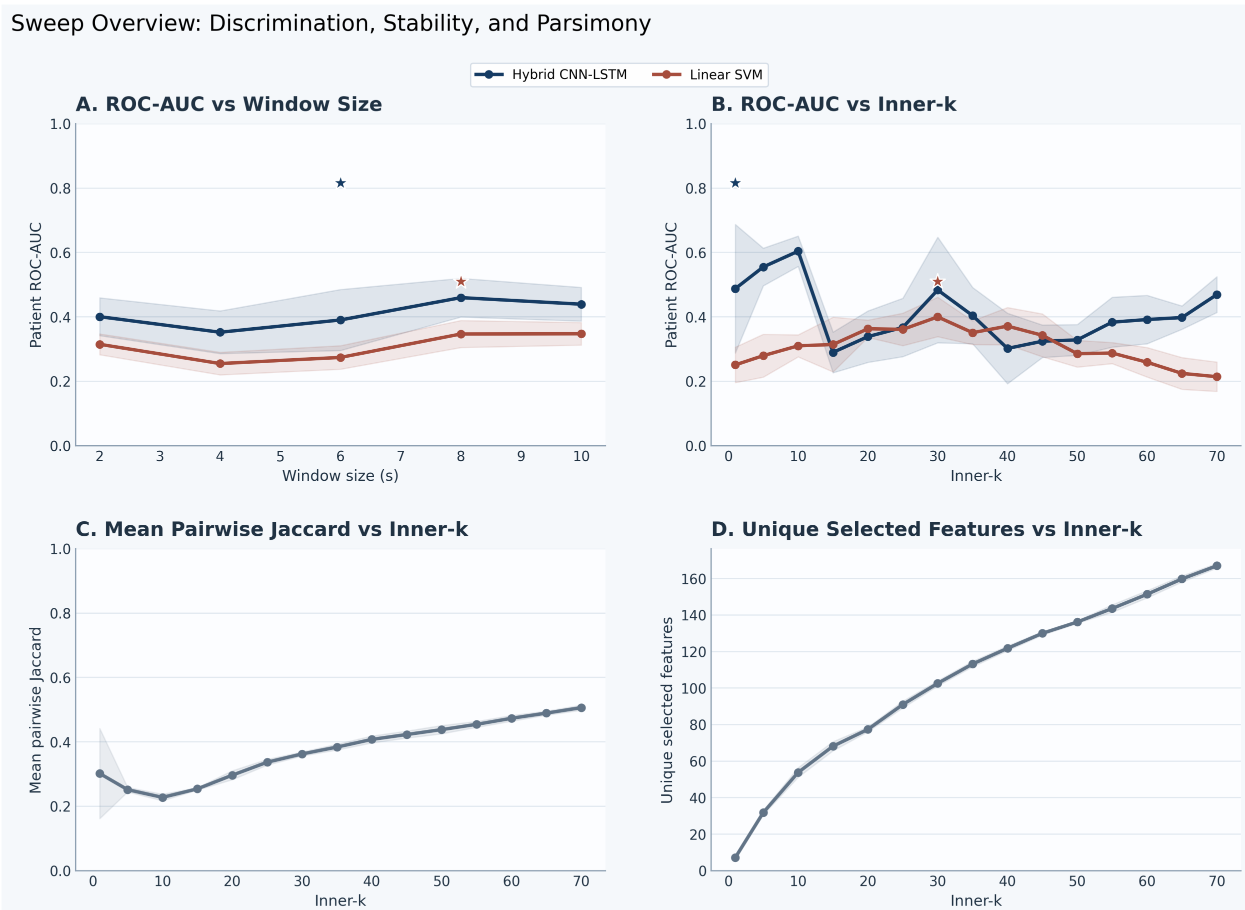

Figure 1: Sweep-level overview across 150 successful runs. The top row compares patient-level ROC-AUC by model as window size and `inner-k` vary. The bottom row shows the shared feature-selection behavior of the pipeline because both classifiers use the same selected feature subsets for matched sweep settings. Stars mark the best individual hybrid and SVM runs.

3.2 Best-run patient-level performance

The best hybrid configuration achieved patient-level ROC-AUC 0.816, PR-AUC 0.579, balanced accuracy 0.786, accuracy 0.810, F1 0.714, and MCC 0.571 over 21 held-out participants (Figure 2; Table 2). Its ROC-AUC bootstrap interval ranged from 0.600 to 0.990, and the permutation test remained significant ($p = 0.008$), which is notable given the relatively small cohort. The corresponding confusion matrix contained 12 true negatives, 5 true positives, 2 false positives, and 2 false negatives.

The best SVM baseline reached balanced accuracy 0.643 and ROC-AUC 0.510, with a nonsignificant permutation result ($p = 0.469$). This weaker performance came with a much larger per-fold feature budget (`inner-k=30`) and a far larger overall candidate set. The contrast indicates that the sparse hybrid pipeline extracted stronger patient-level discrimination from a much smaller subset.

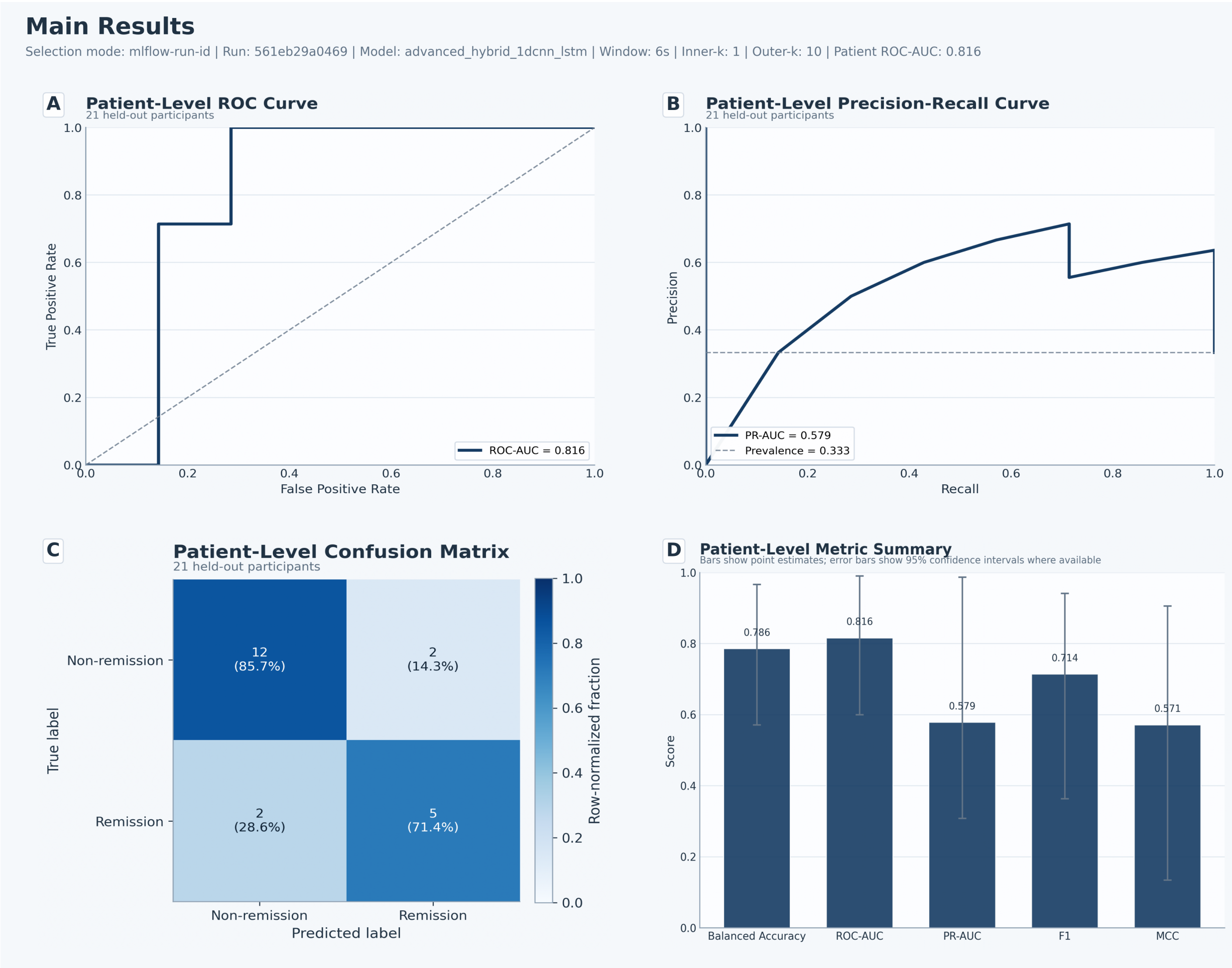

Figure 2: Best-run main-results composite for the hybrid model (`mlflow_run_id=561eb29a046946818320bea18f21bed6`). The figure summarizes the patient-level ROC curve, PR curve, confusion matrix, and metric summary for the highest-performing sweep configuration.

Table 2: Best-performing configuration for each model family.

Model	Window	Inner-k	ROC-AUC	PR-AUC	Bal. Acc.	F1	MCC	Feat./fold	Unique feat.	Mean Jaccard
Hybrid CNN-LSTM	6	1	0.816	0.579	0.786	0.714	0.571	1.0	7	0.190
Linear SVM	8	30	0.510	0.411	0.643	0.533	0.277	30.0	103	0.360

3.3 Sparse biomarker set with modest hard-set overlap

The winning hybrid run selected an average of 1 feature per fold and yielded only 7 unique recurrent features across the entire outer-validation process (Table 3). That degree of sparsity is one of the most distinctive findings in the sweep. Hard-set overlap was modest: mean pairwise Jaccard 0.190 and mean Kuncheva index 0.187. Figure 3 summarizes the small recurrent pool together with the modest chance-adjusted stability summary.

A small recurring feature set emerged repeatedly, though not in every fold, and the selected set was dominated by temporal and frontal-temporal descriptors. The most recurrent feature was `tp10_spectral_entropy`, selected in 8 of 21 folds. The next most recurrent features were `left_frontal_temporal_diff_gamma`, `left_frontal_temporal_diff_beta`, and `af7_zero_crossings`.

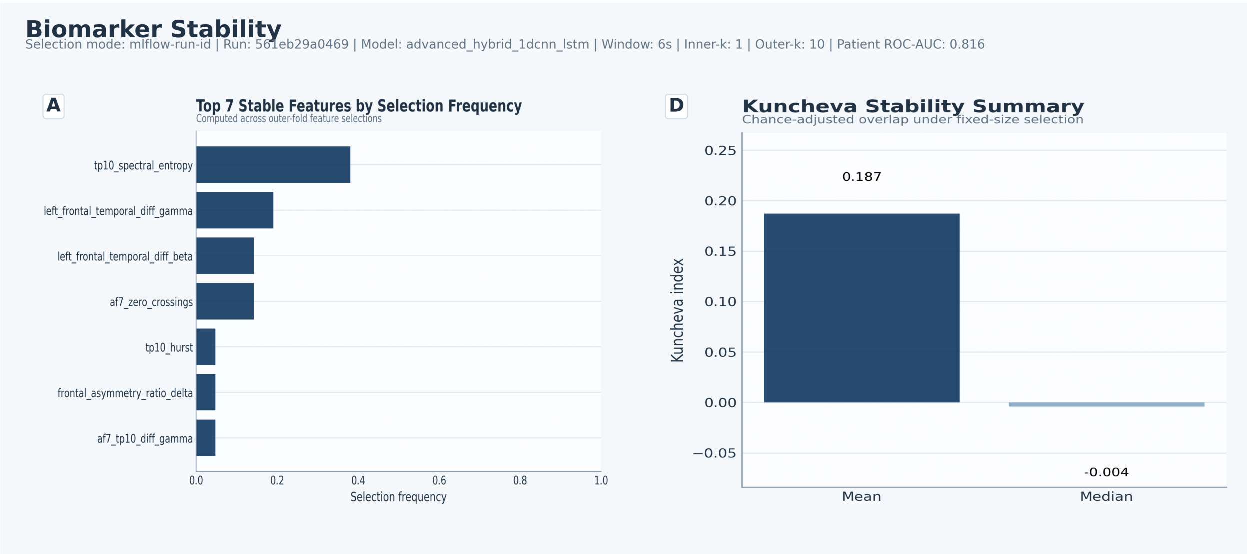

Figure 3: Biomarker-stability composite for the best hybrid run. Panel A ranks the recurrent features by selection frequency. Panel D reports the modest Kuncheva stability summary.

Table 3: Recurrent candidate biomarkers from the best hybrid run. Δ selection is remission minus non-remission selection share.

Feature	Folds	Share	Δ sel.	Sign cons.	Mean d	Effect direction
tp10_spectral_entropy	8	0.381	-0.571	1.000	-0.935	Non-remission > remission
left_frontal_temporal_diff_gamma	4	0.190	-0.071	1.000	-1.059	Non-remission > remission
left_frontal_temporal_diff_beta	3	0.143	0.000	1.000	-0.915	Non-remission > remission
af7_zero_crossings	3	0.143	0.429	1.000	0.814	Remission > non-remission
tp10_hurst	1	0.048	0.143	1.000	-1.010	Non-remission > remission
af7_tp10_diff_gamma	1	0.048	0.143	1.000	-1.203	Non-remission > remission
frontal_asymmetry_ratio_delta	1	0.048	-0.071	1.000	0.794	Remission > non-remission

3.4 Effect-direction consistency and class-conditional patterns

The recurrent hybrid biomarkers showed perfect mean sign consistency across the effect-stability summary. This supports treating them as candidate biomarkers with stable directionality across the folds in which they were selected. `tp10_spectral_entropy` and the frontal-temporal gamma and beta difference terms were repeatedly associated with the non-remission group, whereas `af7_zero_crossings` was more frequently selected in remission folds and retained a positive mean effect size (Figure 4).

This distinction matters clinically. The selection-frequency deltas do not justify definitive mechanistic claims. They provide a concrete interpretation target for follow-up validation. The best hybrid pipeline contributes two linked observations: useful patient-level discrimination can be achieved with an extremely sparse feature budget, and the resulting recurrent candidates show interpretable directionality even when fold-to-fold set overlap remains modest.

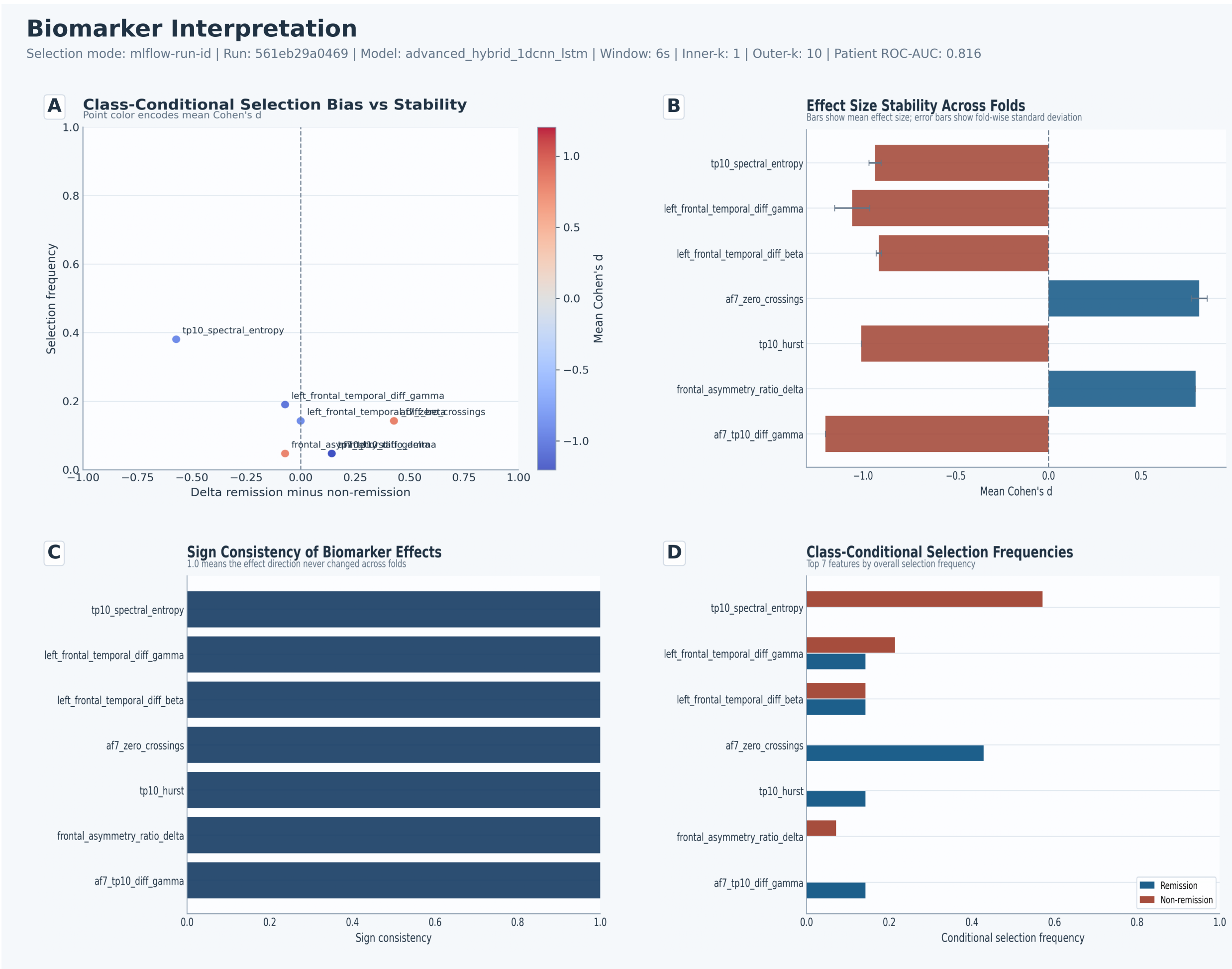

Figure 4: Biomarker-interpretation composite for the best hybrid run. The recurring features show consistent effect direction, with `tp10_spectral_entropy` skewing toward non-remission-associated selections and `af7_zero_crossings` skewing toward remission-associated selections.

4 Discussion

The strongest patient-level discrimination in this sweep emerged from the sparsest tested configuration. The best hybrid run used `inner-k=1`, selected only 7 unique recurrent features overall, and exceeded the best linear baseline by a large margin on ROC-AUC and MCC. In this cohort, performance depended on how effectively the model used a small selected subset.

This result fits part of the EEG depression literature and warrants conservative interpretation. Prior work has argued that EEG can provide clinically useful baseline markers in major depressive disorder and treatment-response settings, and it has also emphasized the need for tighter biomarker discipline, better standardization, and more realistic expectations about generalization (Olbrich and Arns, 2013; Olbrich et al., 2015; Leuchter et al., 2010; Simmatis et al., 2023; Alonso et al., 2017; ?). Machine learning studies have likewise reported encouraging treatment-response signals, including combinations of EEG and clinical covariates, yet translation is often limited by data heterogeneity, cohort size, or insufficient interpretability (?Jaworska et al., 2019; Oakley et al., 2022; Shalbaf et al., 2018; Ebrahimzadeh et al., 2021; ?; ?; ?).

The present sweep adds a feature-selection view to that literature. The shared feature-selection panels in Figure 1 show that larger feature budgets yield more overlap and many more unique discovered features. Those larger budgets do not improve discrimination. The panels support a small recurring candidate set with consistent direction of class separation. That reading keeps the modest Jaccard and Kuncheva scores in view and gives appropriate weight to the stronger effect-direction signal.

The same clinical program has also supported other signal representations in recent colleague papers. PSD-based deep learning on the baseline 4-channel MDD recordings identified informative low-density temporal signal content, and PLV-based analyses from the same home-based trial linked frontal-temporal and temporoparietal synchronization to treatment response (??). In the present sweep, the recurrent candidate set again concentrated in TP10-derived and frontal-temporal descriptors. That regional convergence supports the view that the low-montage recordings contain response-relevant information across multiple feature families.

The specific candidates in the present sweep are plausible enough to deserve follow-up. `tp10_spectral_entropy` was the most frequently selected feature and skewed toward non-remission-associated folds, whereas `af7_zero_crossings` skewed toward remission-associated folds. The recurrent set emphasized temporal and frontal-temporal descriptors, with limited concentration in any single frontal asymmetry family. This regional profile is compatible with the same-program PSD and PLV findings, which also concentrated informative signal in TP10-centered or frontal-temporal relationships (??). The current study contributes a sparse handcrafted candidate set with explicit recurrence and effect-direction summaries.

Related home-based tDCS studies in bipolar depression from the same group point in a similar low-montage direction. PSD-based modeling again highlighted AF7 and TP10 signal content, and PLV analyses emphasized frontal-temporal and temporoparietal synchronization patterns (??). These bipolar studies address a different diagnosis and treatment protocol. They support reduced-montage feasibility and signal plausibility in adjacent affective cohorts.

The dataset is relatively small ($n = 21$), the confidence intervals are wide, and there is no external validation set. The bottom-row sweep panels describe the selection pipeline itself because the same feature-selection layer feeds both classifiers. Class-conditional ratios also become unstable when a feature is absent from one class, so the manuscript uses deltas and effect direction as the main descriptive summaries. The recurring features are best treated as candidate biomarkers for follow-up validation.

Future work should test whether the same sparse feature set recurs under independent cohorts, alternative low-density montages, and explicit external validation (?). Adjacent response-prediction work in depression and neuromodulation provides useful targets for that next stage, including EEG-based prediction of antidepressant response, treatment-resistant depression biomarkers, and neuromodulation-specific response signatures (Funk and George, 2008; Bailey et al., 2023; Guo et al., 2024; ?; ?; ?; Romero-Marín et al., 2026; Kratter et al., 2026; Waller et al., 2026; Sheen et al., 2024; Zeidabadi and Rashidi, 2025; Reissmann et al., 2026; Ulrich et al., 2025; Pettorruso et al., 2024). These extensions should keep stringent patient-level evaluation and sparse interpretable feature budgets in the core study design.

5 Conclusion

In this sweep, low-montage resting-state EEG supported useful patient-level discrimination of tDCS response in adults with treatment-resistant depression, and the strongest result came from an extremely sparse feature-selection regime. The best hybrid run achieved ROC-AUC 0.816 while selecting only 1 feature per fold and 7 unique recurrent features overall. Hard-set overlap across folds remained modest. The present evidence supports a small, interpretable candidate set with consistent effect direction under strict participant-level validation.

That combination of parsimony, interpretability, and subject-independent evaluation defines the main contribution of the present draft. External validation in larger cohorts is the next requirement. The current results support the feasibility of sparse short-timescale EEG biomarker discovery for tDCS response in a clinically realistic low-density setting.

References

Abbott, Christopher C, Calhoun, Vince D, Espinoza, Randall, Jiang, Rongtao, Jones, Tom, Narr, Katherine L, Qi, Shile, Sui, Jing, Sun, Hailun, Upston, Joel, Wade, and Benjamin Sc. Preliminary prediction of individual response to electroconvulsive therapy using whole-brain functional magnetic resonance imaging data. 2020. doi: 10.1016/j.nicl.2019.102080. URL https://escholarship.org/uc/item/12x366xr.

Abelson, J, Akeman, E, Aupperle, Robin L, Bodurka, J, Clausen, AN, Cosgrove, KT, Craske, MG, Kirlic, N, Martell, C, Mathis, B, McDermott, TJ, Paulus, M, Santiago, J, Thompson, WK, Wolitzky-Taylor, and K. Protocol for a randomized controlled trial examining multilevel prediction of response to behavioral activation and exposure-based therapy for generalized anxiety disorder. 2020. doi: 10.21203/rs.2.13364/v2. URL https://escholarship.org/uc/item/6cb012mk.

Agnetti, Virgilio, Aiello, Elena, De Natale, Edoardo, Deriu, Franca, Paulus, Kai Stephan, Sotgiu, Giovanni, Tolu, and Eusebio. Effects of dance therapy on balance, gait and neuro-psychological performances in patients with parkinson's disease and postural instability, 2012. URL https://core.ac.uk/download/16749566.pdf.

Golnoush Alamian, Ana-Sofía Hincapié, Etienne Combrisson, Thomas Thiery, Véronique Martel, Dmitrii Althukov, and Karim Jerbi. Alterations of intrinsic brain connectivity patterns in depression and bipolar disorders: A critical assessment of magnetoencephalography-based evidence. *Frontiers in Psychiatry*, 2017. doi: 10.3389/fpsyt.2017.00041. URL https://doi.org/10.3389/fpsyt.2017.00041.

Lindsay Alexander, Jasmine Escalera, Lei Ai, Charissa Andreotti, Karina Febre, Alexander Mangone, Natan Vega Potler, Nicolas Langer, Alexis Alexander, Meagan Kovacs, Shannon Litke, Bridget O'Hagan, J. E. Andersen, Batya Bronstein, Anastasia Bui, Marijayne Bushey, H. W. Butler, Victoria Castagna, Nicolas L. Camacho, Elisha Chan, Danielle Citera, Jon Clucas, Samantha Cohen, Sarah Dufek, Megan Eaves, Brian Fradera, Judith Gardner, Natalie Grant-Villegas, Gabriella Green, C. Jane Gregory, Emily Hart, Shana Harris, Megan K. Horton, Danielle Kahn, Katherine E. Kabotyanski, Bernard Z. Karmel, Simon P. Kelly, Kayla Kleinman, Bonhwang Koo, Eliza Kramer, Elizabeth M. Lennon, Catherine Lord, Ginny Mantello, Amy Margolis, Kathleen R. Merikangas, Judith Milham, Giuseppe Minniti, Rebecca Neuhaus, Alexandra M. Levine, Yael Osman, Lucas C. Parra, Ken Pugh, Amy Racanello, Anita Restrepo, Tian Saltzman, Batya Septimus, Russell H. Tobe, Rachel Waltz, Anna Williams, Anna J. Yeo, F. Xavier

Castellanos, Arno Klein, Tomáš Paus, Bennett L. Leventhal, R. Cameron Craddock, Harold S. Koplewicz, and Michael P. Milham. An open resource for transdiagnostic research in pediatric mental health and learning disorders. *Scientific Data*, 2017. doi: 10.1038/sdata.2017.181. URL https://doi.org/10.1038/sdata.2017.181.

Elena A. Allen, Eswar Damaraju, Sergey Plis, Erik B. Erhardt, Tom Eichele, and Vince D. Calhoun. Tracking whole-brain connectivity dynamics in the resting state. *Cerebral Cortex*, 2012. doi: 10.1093/cercor/bhs352. URL https://doi.org/10.1093/cercor/bhs352.

Alonso, Esther, Arnott, Stephen R., Atluri, Sravya, Blumberger, Daniel, Brenner, Colleen A., Daskalakis, Zafiris J., Dhami, Prabhjot, Dharsee, Moyez, Evans, Kenneth R., Farzan, Faranak, Frehlich, Matthew, Frey, Benicio N., Kennedy, Sidney H., Kleffner, Killian, Lam, Raymond W., Liotti, Mario, Mcandrews, Mary Pat, Milev, Roumen, Price, Rae, Ravindran, Arun, Rotzinger, Susan, Vila-Rodriguez, Fidel, Wong, and Willy. Standardization of electroencephalography for multi-site, multi-platform and multi-investigator studies: Insights from the canadian biomarker integration network in depression. *Scientific Reports*, 2017. doi: 10.1038/s41598-017-07613-x. URL https://www.research.unipd.it/bitstream/11577/3260320/1/Liotti_SciRep2017.pdf.

BABILONI, CLAUDIO, Boeijinga PH, Brunovsky M, Drinkenburg WH, Ffytche DH, Freeman J, Hegerl U, Hirata K, IPEG Pharmaco EEG Guidelines C.o.m.m.i.t.t.e.e. Collaborators, Jobert M, Kinoshita T, Knott VJ, Lopes Da Silva FH, Matousek M, Mucci A, Nottage JF, Olbrich S, Prichep LS, Ruigt GS, Saletu B, Stancak A, Strik WK, van Gerven JM, Wilson FJ, and Wise RG. Guidelines for the recording and evaluation of pharmaco-eeg data in man: the international pharmaco-eeg society (ipeg). 2012. doi: 10.1159/000343478. URL https://iris.uniroma1.it/bitstream/11573/802874/1/Jobert_Guidelines_2012.pdf.

Ganesh M. Babulal, Yakeel T. Quiroz, Benedict C. Albensi, Eider M. Arenaza-Urquijo, Arlene Astell, Claudio Babiloni, Alex Bahar-Fuchs, J. Simon Bell, Gene L. Bowman, Adam M. Brickman, Gaël Chételat, Carrie Ciro, Ann D. Cohen, Peggye Dilworth-Anderson, Hiroko H. Dodge, Simone Dreux, Steven D. Edland, Anna J. Esbensen, Lisbeth Evered, Michael Ewers, Keith N. Fargo, Juan Fortea, Hector M. González, Deborah Gustafson, Elizabeth Head, James A. Hendrix, Scott M. Hofer, Leigh Johnson, Roos J. Jutten, Kerry Kilborn, Krista L. Lanctôt, Jennifer J. Manly, Ralph N. Martins, Michelle M. Mielke, Martha Clare Morris, Melissa E. Murray, Esther S. Oh, Mario A. Parra, Robert A. Rissman, Catherine M. Roe, Octavio A. Santos, Nikolaos Scarmeas, Lon S. Schneider, Nicole Schupf, Sietske A.M. Sikkes, Heather M. Snyder, Hamid R. Sohrabi, Yaakov Stern, André Strydom, Yi Tang, Graciela Muñiz-Terrera, Charlotte E. Teunissen, Debora Melo van Lent, Michael Weinborn, Linda M.P. Wesselman, Donna M. Wilcock, Henrik Zetterberg, and Sid E. O'Bryant. Perspectives on ethnic and racial disparities in alzheimer's disease and related dementias: Update and areas of immediate need. *Alzheimer s & Dementia*, 2018. doi: 10.1016/j.jalz.2018.09.009. URL https://doi.org/10.1016/j.jalz.2018.09.009.

N. Bailey, K. Hoy, C. Sullivan, B. Allman, N. Rogasch, Z. Daskalakis, and P. Fitzgerald. Concurrent transcranial magnetic stimulation and electroencephalography measures are associated with antidepressant response from rtms treatment for depression. *medRxiv*, 2023. doi: 10.1101/2023.02.10.23285794. URL https://www.semanticscholar.org/paper/8278c0b8d2b48ae5ed473c09d6c3f9ba34f53391.

Paulo S. Boggio, Sergio P. Rigonatti, Rafael Bernardon Ribeiro, Martin Luiz Myczkowski, Michael A. Nitsche, Álvaro Pascual-Leone, and Felipe Fregni. A randomized, double-blind clinical trial on the efficacy of cortical direct current stimulation for the treatment of major depression.

The International Journal of Neuropsychopharmacology, 2007. doi: 10.1017/s1461145707007833. URL https://doi.org/10.1017/s1461145707007833.

Alexander A. Borbély, Serge Daan, Anna Wirz-Justice, and Tom Deboer. The two-process model of sleep regulation: a reappraisal. *Journal of Sleep Research*, 2016. doi: 10.1111/jsr.12371. URL https://doi.org/10.1111/jsr.12371.

Cecchi, Nicholas J, Gerges, Paul, Hicks, James W, Monroe, Derek C, Phreaner, Jenna, Small, and Steven L. A dose relationship between brain functional connectivity and cumulative head impact exposure in collegiate water polo players. 2020. doi: 10.3389/fneur.2020.00218. URL https://escholarship.org/content/qt3dk7m0zg/qt3dk7m0zg.pdf?t=qafdxa.

Wei-Liang Chen, Julie C. Wagner, Nicholas Heugel, Jeffrey Sugar, Yu-Wen Lee, Lisa L. Conant, Marsha Malloy, Joseph Heffernan, Brendan J. Quirk, Anthony Zinos, Scott A. Beardsley, Robert W. Prost, and Harry T. Whelan. Functional near-infrared spectroscopy and its clinical application in the field of neuroscience: Advances and future directions. *Frontiers in Neuroscience*, 2020. doi: 10.3389/fnins.2020.00724. URL https://doi.org/10.3389/fnins.2020.00724.

Steven C. Cramer, Mriganka Sur, Bruce H. Dobkin, Charles P. O'Brien, Terence D. Sanger, John Q. Trojanowski, Judith M. Rumsey, Ramona Hicks, Judy L. Cameron, David Chen, Wen Chen, Leonardo G. Cohen, C. deCharms, C. J. Duffy, Guinevere F. Eden, E. E. Fetz, Rosemarie Filart, M C Freund, Steven Grant, Suzanne N. Haber, Peter W. Kalivas, Bryan Kolb, Arthur F. Kramer, Michael P. Lynch, Helen S. Mayberg, Patrick S. McQuillen, Ralph Nitkin, A. Pascual–Leone, Patricia A. Reuter-Lorenz, Nicholas D. Schiff, Anu Sharma, L. Shekim, Michael P. Stryker, Edith V. Sullivan, and Sophia Vinogradov. Harnessing neuroplasticity for clinical applications. *Brain*, 2011. doi: 10.1093/brain/awr039. URL https://doi.org/10.1093/brain/awr039.

Karen D. Davis, Nima Aghaeepour, Andrew H. Ahn, Martin S. Angst, David Borsook, Ashley Brenton, Michael E. Burczynski, Christopher Crean, Robert R. Edwards, Brice Gaudillière, Georgene W. Hergenroeder, Michael J. Iadarola, Smriti Iyengar, Yunyun Jiang, Jiang-Ti Kong, Sean Mackey, Carl Y. Saab, Christine N. Sang, Joachim Scholz, Märta Segerdahl, Irene Tracey, Christin Veasley, Jing Wang, Tor D. Wager, Ajay D. Wasan, and Mary Ann Pelleymounter. Discovery and validation of biomarkers to aid the development of safe and effective pain therapeutics: challenges and opportunities. *Nature Reviews Neurology*, 2020. doi: 10.1038/s41582-020-0362-2. URL https://doi.org/10.1038/s41582-020-0362-2.

Hanneke van Dijk, Guido van Wingen, Damiaan Denys, Sebastian Olbrich, Rosalinde van Ruth, and Martijn Arns. The two decades brainclinics research archive for insights in neurophysiology (tdbrain) database. *Scientific Data*, 2022. doi: 10.1038/s41597-022-01409-z. URL https://doi.org/10.1038/s41597-022-01409-z.

Duncan, John S, Elger, Christian E, Engel, Jerome, Staba, Richard, Vakharia, Vejay N, Witt, and Juri-Alexander. Getting the best outcomes from epilepsy surgery., 2018. URL https://escholarship.org/content/qt0120f648/qt0120f648.pdf?t=qaehok.

Wallace C. Duncan, Simone Sarasso, Fabio Ferrarelli, Jessica Selter, Brady A. Riedner, Nadia S. Hejazi, Peixiong Yuan, Nancy E. Brutsché, Husseini K. Manji, Giulio Tononi, and Carlos A. Zarate. Concomitant bdnf and sleep slow wave changes indicate ketamine-induced plasticity in major depressive disorder. *The International Journal of Neuropsychopharmacology*, 2012. doi: 10.1017/s1461145712000545. URL https://doi.org/10.1017/s1461145712000545.

Elias Ebrahimzadeh, Mostafa Asgarinejad, Sarah Saliminia, Sarvenaz Ashoori, and Masoud Seraji. Predicting clinical response to transcranial magnetic stimulation in major depression using time-frequency eeg signal processing. *Biomedical Engineering Applications Basis and Communications*, 2021. doi: 10.4015/s1016237221500484. URL https://doi.org/10.4015/s1016237221500484.

Agnes P. Funk and Mark S. George. Prefrontal eeg asymmetry as a potential biomarker of antidepressant treatment response with transcranial magnetic stimulation (tms): A case series. *Clinical EEG and Neuroscience*, 2008. doi: 10.1177/155005940803900306. URL https://doi.org/10.1177/155005940803900306.

Xiaotong Gu, Zehong Cao, Alireza Jolfaei, Peng Xu, Dongrui Wu, Tzyy-Ping Jung, and Chin-Teng Lin. Eeg-based brain-computer interfaces (bcis): A survey of recent studies on signal sensing technologies and computational intelligence approaches and their applications. *UTS ePRESS (University of Technology Sydney)*, 2021. doi: 10.1109/tcbb.2021.3052811. URL http://hdl.handle.net/10453/147196.

Ling Guo, Zhuo Zhang, X. Tan, K. Phua, C. Wang, P. Tor, and K. Ang. Resting-state eeg biomarkers of accelerated intermittent theta burst stimulation treatment for depression: a pilot study. In *Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, 2024. doi: 10.1109/EMBC53108.2024.10782112. URL https://www.semanticscholar.org/paper/9d00f9320f2f75435719d9cccc4133e989f8737e.

Christoph S. Herrmann, Stefan Rach, Toralf Neuling, and Daniel Strüber. Transcranial alternating current stimulation: a review of the underlying mechanisms and modulation of cognitive processes. *Frontiers in Human Neuroscience*, 2013. doi: 10.3389/fnhum.2013.00279. URL https://doi.org/10.3389/fnhum.2013.00279.

Natalia Jaworska, Sara de la Salle, Mohamed Hamza Ibrahim, Pierre Blier, and Verner Knott. Leveraging machine learning approaches for predicting antidepressant treatment response using electroencephalography (eeg) and clinical data. *Frontiers in Psychiatry*, 2019. doi: 10.3389/fpsyt.2018.00768. URL https://doi.org/10.3389/fpsyt.2018.00768.

Daniel Keeser, Thomas Meindl, Julie Bor, Ulrich Palm, Oliver Pogarell, Christoph Mulert, Jérôme Brunelin, Hans-Jürgen Möller, Maximilian F. Reiser, and Frank Padberg. Prefrontal transcranial direct current stimulation changes connectivity of resting-state networks during fmri. *Journal of Neuroscience*, 2011. doi: 10.1523/jneurosci.0542-11.2011. URL https://doi.org/10.1523/jneurosci.0542-11.2011.

Weizhuang Kong, Zhe Sun, Jing Zhu, Lingjiang Li, Guanru Wang, Xuexiao Shao, Xiaowei Li, and Bin Hu. Alterations in temporal-spatial brain entropy in treatment-resistant depression treated with nitrous oxide: Evidence from resting-state eeg. *Clinical Neurophysiology*, 2025. doi: 10.1016/j.clinph.2025.01.014. URL https://www.semanticscholar.org/paper/4a2e72d2cd526876793bda71fe0052e275f6fec2.

Ian H. Kratter, Christopher W. Austelle, Jennifer I. Lissemore, Masataka Wada, Andrew Geoly, Anna Chaiken, Irakli Kaloiani, Noriah Johnson, Stephanie Wan, Lena Kozyr, Ethan Makarewycz, Brendan L. Wong, Malvika Sridhar, Flint M. Espil, Nick Bassano, Bora Kim, Jarrod Ehrie, Adi Maron-Katz, Claudia Tischler, Romina Nejad, Jean-Marie Batail, Angela Phillips, Eleanor Cole, Tiffany J. Ford, Brandon S. Bentzley, Booil Jo, Alan F. Schatzberg, David Spiegel, Cammie Rolle, Gregory L. Sahlem, and Nolan Williams. Stanford neuromodulation therapy for treatment-resistant depression: a randomized controlled trial confirming efficacy, and an <scp>eeg</scp>

study providing insight into mechanism of action and a potentially predictive biomarker of efficacy. *World Psychiatry*, 2026. doi: 10.1002/wps.70032. URL https://doi.org/10.1002/wps.70032.

Jean-Pascal Lefaucheur, André Alemán, Chris Baeken, David Benninger, Jérôme Brunelin, Vincenzo Di Lazzaro, Saša R. Filipović, Christian Grefkes, Alkomiet Hasan, Friedhelm C. Hummel, Satu K. Jääskeläinen, Berthold Langguth, Letizia Leocani, Alain Londero, Raffaele Nardone, Jean-Paul Nguyen, Thomas Nyffeler, Albino J. Oliveira-Maia, Antonio Oliviero, Frank Padberg, Ulrich Palm, Walter Paulus, Emmanuel Poulet, Angelo Quartarone, Fady Rachid, Irena Rektorová, Símone Rossi, Hanna Sahlsten, Martin Schecklmann, David Szekely, and Ulf Ziemann. Evidence-based guidelines on the therapeutic use of repetitive transcranial magnetic stimulation (rtms): An update (2014–2018). *Clinical Neurophysiology*, 2020. doi: 10.1016/j.clinph.2019.11.002. URL https://doi.org/10.1016/j.clinph.2019.11.002.

Andrew F. Leuchter, Ian A. Cook, Steven P. Hamilton, Katherine L. Narr, Arthur Toga, Aimee M. Hunter, Kym Faull, Julian Whitelegge, Anne M. Andrews, Joseph Loo, Baldwin Way, Stanley F. Nelson, Steven Horvath, and Barry D. Lebowitz. Biomarkers to predict antidepressant response. *Current Psychiatry Reports*, 2010. doi: 10.1007/s11920-010-0160-4. URL file:///data/remote/core/dit/data/Springer-OA/pdf/376/aHR0cDovL2xpbmsuc3ByaW5nZXIuY29tLzEwLjEwMDcvczExOTIwLTAxMC0wMTYwLTQucGRm.pdf.

Qiang Li, Michael J. Detke, Steve Paul, William Z. Potter, Fan Zhang, Alan Breier, Larry Alphs, Owen M. Wolkowitz, Larry Ereshefsky, Gregory G. Grecco, and Ken Wang. Machine learning-enabled eeg biomarkers predict divergent antidepressant and placebo response in a clinical trial of major depression. 2025. doi: 10.1101/2025.05.29.25328167. URL https://doi.org/10.1101/2025.05.29.25328167.

Ian G. McKeith, Bradley F. Boeve, Dennis W. Dickson, Glenda M. Halliday, John-Paul Taylor, Daniel Weintraub, Dag Aarsland, James E. Galvin, Johannes Attems, Clive Ballard, Ashley Bayston, Thomas G. Beach, Frederic Blanc, Nicolaas I. Bohnen, Laura Bonanni, José Brás, Patrik Brundin, David J. Burn, Alice Chen-Plotkin, John E. Duda, Omar M. A. El-Agnaf, Howard Feldman, Tanis J. Ferman, Dominic ffytche, Hiroshige Fujishiro, Douglas Galasko, Jennifer G. Goldman, Stephen N. Gomperts, Neill R. Graff-Radford, Lawrence S. Honig, Álex Iranzo, Kejal Kantarci, Daniel Kaufer, Walter A. Kukull, Virginia M.-Y. Lee, James B. Leverenz, Simon J.G. Lewis, Carol F. Lippa, Angela Lunde, Mario Masellis, Eliezer Masliah, Pamela J. McLean, Brit Mollenhauer, Thomas J. Montine, Emilio Moreno, Etsuro Mori, Melissa E. Murray, John T. O'Brien, Sotoshi Orimo, Ronald B. Postuma, Shankar Ramaswamy, Owen A. Ross, David P. Salmon, Andrew Singleton, Angela Taylor, Alan Thomas, Pietro Tiraboschi, Jon B. Toledo, John Q. Trojanowski, Debby W. Tsuang, Zuzana Walker, Masahito Yamada, and Kenji Kosaka. Diagnosis and management of dementia with lewy bodies. *Neurology*, 2017. doi: 10.1212/wnl.0000000000004058. URL https://doi.org/10.1212/wnl.0000000000004058.

Roumen Milev, Peter Giacobbe, Sidney H. Kennedy, Daniel M. Blumberger, Zafiris J. Daskalakis, Jonathan Downar, Mandana Modirrousta, Simon Patry, Fidel Vila-Rodriguez, Raymond W. Lam, Glenda MacQueen, Sagar V. Parikh, Arun Ravindran, and the CANMAT Depression Work Group. Canadian network for mood and anxiety treatments (canmat) 2016 clinical guidelines for the management of adults with major depressive disorder. *The Canadian Journal of Psychiatry*, 2016. doi: 10.1177/0706743716660033. URL https://doi.org/10.1177/0706743716660033.

Nicholas Murphy, Amanda J. F. Tamman, Marijn Lijffijt, Dania Amarneh, Sidra Iqbal, Alan C. Swann, Lynnette A. Averill, Brittany O'Brien, and Sanjay J. Mathew. Neural complexity eeg biomarkers of rapid and post-rapid ketamine effects in late-life treatment-resistant depression: a randomized control trial. *Neuropsychopharmacology*, 2023. doi: 10.1038/s41386-023-01586-4. URL https://doi.org/10.1038/s41386-023-01586-4.

Thomas Oakley, Jonathan Coskuner, Andrew Cadwallader, Maryam Ravan, and Gary Hasey. Eeg biomarkers to predict response to sertraline and placebo treatment in major depressive disorder. *IEEE Transactions on Biomedical Engineering*, 2022. doi: 10.1109/tbme.2022.3204861. URL https://doi.org/10.1109/tbme.2022.3204861.

Sebastian Olbrich and Martijn Arns. Eeg biomarkers in major depressive disorder: Discriminative power and prediction of treatment response. *International Review of Psychiatry*, 2013. doi: 10.3109/09540261.2013.816269. URL https://doi.org/10.3109/09540261.2013.816269.

Sebastian Olbrich, Rik van Dinteren, and Martijn Arns. Personalized medicine: Review and perspectives of promising baseline eeg biomarkers in major depressive disorder and attention deficit hyperactivity disorder. *Neuropsychobiology*, 2015. doi: 10.1159/000437435. URL https://doi.org/10.1159/000437435.

Ahmet Omurtag, Delpy D T, Durduran T, Gudiño-Mendoza B, Haleh Aghajani, Hasan Onur Keles, Schudlo L C, Wickens C D, and Zander T O. Decoding human mental states by whole-head eeg+fnirs during category fluency task performance. 2017. doi: 10.1088/1741-2552/aa814b. URL http://irep.ntu.ac.uk/id/eprint/32726/1/PubSub10246_799a_Omurtag.pdf.

Sarah K. Peters, Katharine Dunlop, and Jonathan Downar. Cortico-striatal-thalamic loop circuits of the salience network: A central pathway in psychiatric disease and treatment. *Frontiers in Systems Neuroscience*, 2016. doi: 10.3389/fnsys.2016.00104. URL https://doi.org/10.3389/fnsys.2016.00104.

M. Pettorruso, Giorgio Di Lorenzo, Beatrice Benatti, G. d'Andrea, C. Cavallotto, Rosalba Carullo, G. Mancusi, O. Di Marco, G. Mammarella, A. D'attilio, Elisabetta Barlocci, Ilenia Rosa, Alessio Cocco, Lorenzo Pio Padula, Giovanna Bubbico, M. G. Perrucci, Roberto Guidotti, A. D'Andrea, L. Marzetti, F. Zoratto, B. Dell'osso, and Giovanni Martinotti. Overcoming treatment-resistant depression with machine-learning based tools: a study protocol combining eeg and clinical data to personalize glutamatergic and brain stimulation interventions (selectool project). *Frontiers in Psychiatry*, 2024. doi: 10.3389/fpsyt.2024.1436006. URL https://www.semanticscholar.org/paper/169e53dbe54745e1df0495183343bda9d6a8a29e.

Björn Rasch and Jan Born. About sleep's role in memory. *Physiological Reviews*, 2013. doi: 10.1152/physrev.00032.2012. URL https://doi.org/10.1152/physrev.00032.2012.

Andreas Reissmann, Maximilian Rupprecht, B. Langguth, Johanna Rischer, and S. Schoisswohl. Theta-cordance as a biomarker of treatment response to intermittent theta burst stimulation in patients with treatment-resistant depression. *Clinical Neurophysiology*, 2026. doi: 10.1016/j.clinph.2026.2111710. URL https://www.semanticscholar.org/paper/c2c05fd78b73d7ebd82bcbd5cb8af3516e1f5751.

R. Romero-Marín, S. López-Rodríguez, S. Lakis-Granell, E. Buloz-Osorio, M. Cabello-Toscano, Mikel Urretavizcaya-Sarachaga, Maria del Pino Alonso-Ortega, Mohit Chopra, J. Solana-Sánchez, J. Camprodon, Á. Pascual-Leone, D. Bartrés-Faz, Davide Cappon, and G. Cattaneo. Remotely supervised home-based tdcs in treatment-resistant depression: Feasibility, effectiveness and eeg biomarkers of response. *Research Square*, 2026. doi: 10.21203/rs.3.rs-8709698/v1. URL https://www.semanticscholar.org/paper/7ba628353818d20f18ac5ecdccca2bd39bca3d77.

Símone Rossi, Andrea Antal, Sven Bestmann, Marom Bikson, Carmen C. Brewer, Jürgen Brockmöller, Linda L. Carpenter, M. Cincotta, Robert Chen, Jeff Daskalakis, Vincenzo Di Lazzaro, Michael Fox, Mark S. George, Donald L. Gilbert, Vasilios . Kimiskidis, Giacomo Koch, Risto J. Ilmoniemi, Jean Pascal Lefaucheur, Letizia Leocani, Sarah H. Lisanby, Carlo Miniussi, Frank Padberg, Álvaro Pascual-Leone, Walter Paulus, Angel V. Peterchev, Angelo Quartarone, Alexander Rotenberg, John C. Rothwell, Paolo Maria Rossini, Emiliano Santarnecchi, Mouhsin M. Shafi, Hartwig R. Siebner, Yoshikatzu Ugawa, Eric M. Wassermann, Abraham Zangen, Ulf Ziemann, and Mark Hallett. Safety and recommendations for tms use in healthy subjects and patient populations, with updates on training, ethical and regulatory issues: Expert guidelines. *Clinical Neurophysiology*, 2020. doi: 10.1016/j.clinph.2020.10.003. URL https://doi.org/10.1016/j.clinph.2020.10.003.

Reza Shalbaf, Colleen A. Brenner, C. Pang, Daniel M. Blumberger, Jonathan Downar, Zafiris J. Daskalakis, Joseph Tham, Raymond W. Lam, Faranak Farzan, and Fidel Vila-Rodriguez. Nonlinear entropy analysis in eeg to predict treatment response to repetitive transcranial magnetic stimulation in depression. *Frontiers in Pharmacology*, 2018. doi: 10.3389/fphar.2018.01188. URL https://doi.org/10.3389/fphar.2018.01188.

J. Sheen, F. Mazza, D. Momi, J. Miron, Farrokh Mansouri, Thomas Russell, Ryan Zhou, M. Hyde, L. Fox, Helena Voetterl, E. B. Assi, Z. Daskalakis, D. Blumberger, John D. Griffiths, and Jonathan Downar. N100 as a response prediction biomarker for accelerated 1hz right dlpfc-rtms in major depression. *Journal of Affective Disorders*, 2024. doi: 10.1016/j.jad.2024.07.131. URL https://www.semanticscholar.org/paper/c0dabdc70010a5fb8ea6f2546d37fe9e83a13071.

Leif Simmatis, Emma E. Russo, Joseph Geraci, Irene E. Harmsen, and Nardin Samuel. Technical and clinical considerations for electroencephalography-based biomarkers for major depressive disorder. *npj Mental Health Research*, 2023. doi: 10.1038/s44184-023-00038-7. URL https://doi.org/10.1038/s44184-023-00038-7.

Axel Steiger and Marcel Pawlowski. Depression and sleep. *International Journal of Molecular Sciences*, 2019. doi: 10.3390/ijms20030607. URL https://doi.org/10.3390/ijms20030607.

Axel Steiger, Marcel Pawlowski, and Mayumi Kimura. Sleep electroencephalography as a biomarker in depression. *ChronoPhysiology and Therapy*, 2015. doi: 10.2147/cpt.s41760. URL https://doi.org/10.2147/cpt.s41760.

Hayley Thair, Amy L. Holloway, Roger Newport, and Alastair D. Smith. Transcranial direct current stimulation (tdcs): A beginner's guide for design and implementation. *Frontiers in Neuroscience*, 2017. doi: 10.3389/fnins.2017.00641. URL https://doi.org/10.3389/fnins.2017.00641.

Yi-Chun Tsai, Cheng-Ta Li, and Chi-Hung Juan. A review of critical brain oscillations in depression and the efficacy of transcranial magnetic stimulation treatment. *Frontiers in Psychiatry*, 2023. doi: 10.3389/fpsyt.2023.1073984. URL https://doi.org/10.3389/fpsyt.2023.1073984.

Sarah Ulrich, Else Schneider, Gunnar Deuring, Saskia Erni, Maria de Ridder, Jan Sarlon, and Annette Beatrix Brühl. Alterations in resting-state eeg functional connectivity in patients with major depressive disorder receiving electroconvulsive therapy: A systematic review. *Neuroscience & Biobehavioral Reviews*, 2025. doi: 10.1016/j.neubiorev.2025.106017. URL https://doi.org/10.1016/j.neubiorev.2025.106017.

Darcy A. Waller, Linda L. Carpenter, and Stephanie R. Jones. Transient frontal spectral events from eeg predict antidepressant response to sertraline in depression. *Journal of Psychiatric Research*, 2026. doi: 10.1016/j.jpsychires.2026.03.010. URL https://www.semanticscholar.org/paper/6a3e902c05c07569afa93b0bb3e0b484bf519824.

Tino Zaehle, Stefan Rach, and Christoph S. Herrmann. Transcranial alternating current stimulation enhances individual alpha activity in human eeg. *PLoS ONE*, 2010. doi: 10.1371/journal.pone.0013766. URL https://doi.org/10.1371/journal.pone.0013766.

Ali Asadi Zeidabadi and Saeid Rashidi. Prediction of rtms treatment response in depression using a frequency-based eeg biomarker. In *International Conference on Computer and Knowledge Engineering*, 2025. doi: 10.1109/ICCKE68588.2025.11273345. URL https://www.semanticscholar.org/paper/fa7b0c55f517ebfbaa6af5be40ba57129e5d87f2.